

Computing bark from frequency

Bark is a unit used in psychoacoustics which better represents how human listeners perceive pitch. Traunmüller (1990) proposed the following formula to convert frequency in Hertz (Hz) into barks:

$$b' = (26.81 \times f) / (1960 + f) - 0.53$$

where b' is the *uncorrected* value in barks and f is the frequency in Hz.

For very large or small values, a correction is needed:

- If b' is smaller than 2: $b = b' + 0.15 * (2 - b')$
- If b' is larger than 20: $b = b' + 0.22 * (b' - 20.1)$
- Otherwise, $b = b'$

Task

Write a function `hz2bark` which takes a single `int` or `float` value representing a frequency in Hz, and returns the corrected b computed as explained above. Round the result to 3 decimal places using the `round` function.

Put your code in the file `phonetics.py`.

Example

```
In [1]: import phonetics
```

```
In [2]: phonetics.hz2bark(65.4)
```

```
Out[2]: 0.585
```

```
In [3]: phonetics.hz2bark(600)
```

```
Out[3]: 5.754
```

```
In [4]: phonetics.hz2bark(10000)
```

```
Out[4]: 22.279
```

Reading

- PY4E chapters 3 and 4